

HOTEL DEMAND CANCELLATION RATE ANALYSIS

Identifying Cancellation Risk Factors and predicting booking cancellation risk



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Executive Summary

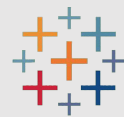
Project Description

Aloha Hotel Group experienced significant booking cancellations during 2015–2017, **37% overall cancellation rate** creates significant demand uncertainty.

Objectives:

- Identify key factors associated with booking cancellations
- Predict cancellation likelihood using logistic regression
- Develop actionable recommendations to reduce cancellation risk

TOOLS



Northstars Metrics

Cancellation Rate

Findings

- **37% overall cancellation rate**
- **99.4% cancellation for non-refundable bookings**
- **56.8% cancellation for 180+ day lead times**
- **94.4% cancellation with 1 previous cancellation**

Recommendation

1. **90-Day Lead-Time Monitoring Rule**
2. **Review TA/TO partner booking & confirmation processes**
3. **Build a Cancellation-Risk Flag System**

Project Background

Company Overview

ALOHA HOTEL GROUP operates on a hybrid B2C and B2B model, catering to both individual customers and corporate clients. Bookings can be made directly through the hotel's channels or via online travel agencies (OTAs), offering flexibility and broad market reach. The primary source of revenue for the group is derived from hotel room stays and lodging services.

Business Challenge

Adapted from Antonio et al. (2017). **Hotel booking cancellations create uncertainty in demand management by reducing forecast accuracy, which can affect pricing, inventory allocation, and overbooking decisions.**

Identifying key factors associated with hotel booking cancellations can help hotels:

- **Improve demand forecasting** and estimate net demand more accurately
- **Optimize pricing and room inventory** based on expected cancellations
- **Develop targeted cancellation policies**
- **Improve overbooking strategies** and reduce empty rooms
- **Reduce demand uncertainty** and support better revenue management

Source: Adapted from Antonio et al. (2017).

Problem Statement & Objective

What factors are associated with hotel booking cancellations between 2015 and 2017?

Objective of Analysis:

1. Identify factors associated with hotel booking cancellations
2. Predict the likelihood of booking cancellation.
3. Provide actionable insights to reduce cancellations

Hypothesis

1. Bookings with a non-refundable deposit type are more likely to be cancelled.
2. Bookings through TA/TO distribution channels are associated with higher cancellation rates .
3. Longer lead time is associated with higher cancellation likelihood

Dataset Overview & Scope of Analysis

Dataset Overview

The dataset provides data at the granularity of **individual bookings/reservations**. Each row contains comprehensive information about a reservation, including customer details, booking characteristics, stay specifics, pricing, and the final reservation status.

The dataset comprises 119,390 bookings across 32 columns. Key data points include booking status, arrival dates, customer characteristics, stay duration, room information, booking channels, cancellation history, and Average Daily Rate (ADR).

Data Source: [Click Here](#)

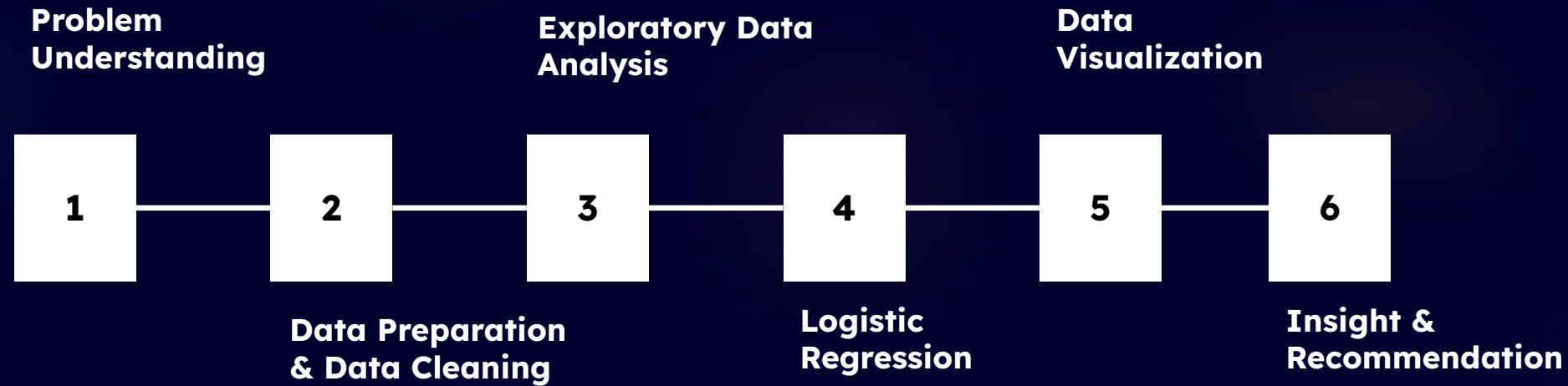
Scope of Analysis

This analysis focus on factors **associated with hotel booking cancellations**, including booking characteristics, customer characteristics, booking channels, hotel type, and seasonal patterns

Dataset Limitation

The analysis does not assess **actual revenue loss, operating costs, profitability, or the direct causes of cancellations** because the dataset does not contain the necessary information.

Methodology



A structured data analysis approach was used to identify key factors associated with hotel booking cancellations, from problem definition and data preparation through exploratory analysis, logistic regression, visualization, and actionable recommendations

Overall Cancellation Rate

Cancellation Rate Indicates a Significant Business Challenge. compared with [SiteMinder's 2025 global hotel benchmark](#) of 19.15%. This is approximately 1.9 times higher, suggesting a significant level of booking uncertainty

Aloha Hotel Cancellation Rate

37%

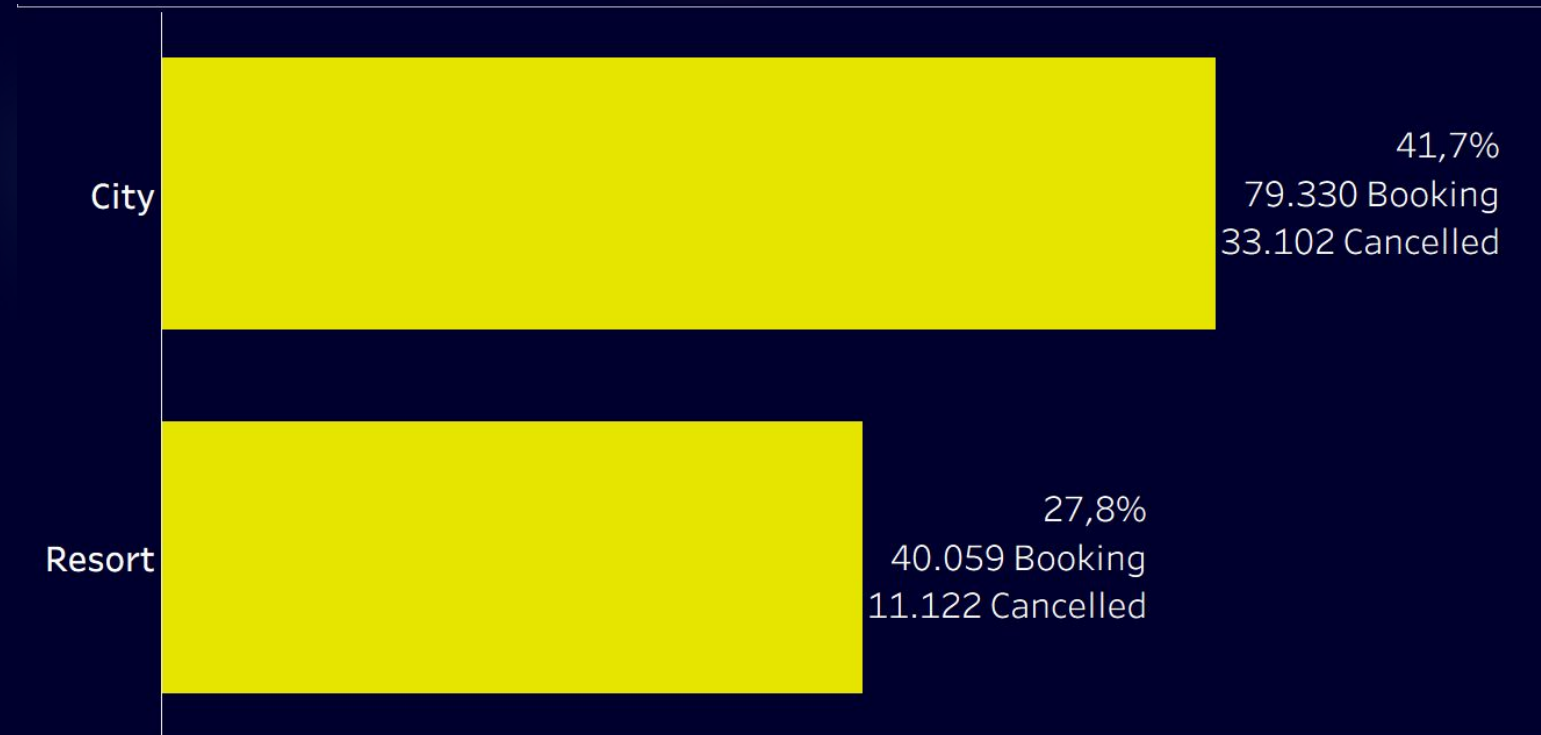
Global Hotel Cancellation Rate (Benchmark)

19%



**+18 percentage points
above benchmark**

Cancellation Rate by Hotel Type



With total bookings records : **119.389**

City Hotel has **higher volume** and shows a **13.9 percentage-point higher cancellation rate** than Resort Hotel, indicating that the difference is not solely driven by booking volume.

Non-Refundable Bookings Contribute to City Hotel's Higher Cancellation Rate

Hotel	No Deposit	Non Refund	Refundable
City	30,5% 66.442 Bookings 20.244 Cancelled	99,8% 12.868 Bookings 12.844 Cancelled	70,0% 20 Bookings 14 Cancelled
Resort	24,7% 38.198 Bookings 9.450 Cancelled	96,0% 1.719 Bookings 1.650 Cancelled	15,5% 142 Bookings 22 Cancelled

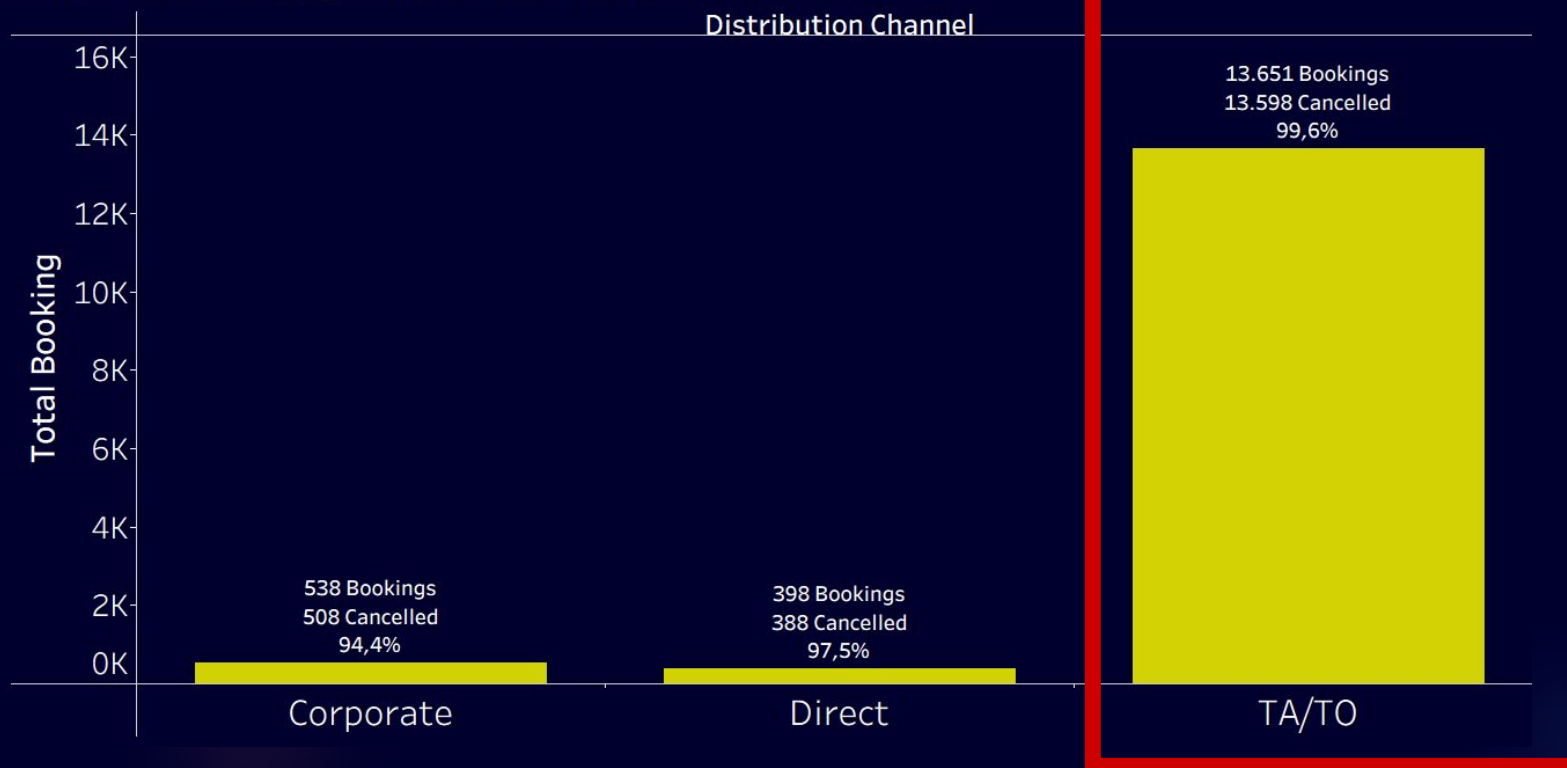
 **Insight:**

Non-refundable bookings show cancellation rates above **96% across both hotel types suggesting non-refund is one of the strong cancellation driver**

City Hotel's higher cancellation rate is partly **driven by its large volume of non-refundable bookings**, which have a **99.8%** cancellation rate.

Non-Refund Bookings Dominated by TA/TO Channel

Non Refund Booking Volume and Cancellation



Non-Refundable Bookings Are Concentrated in TA/TO, with High Cancellation Rates Across All Channels

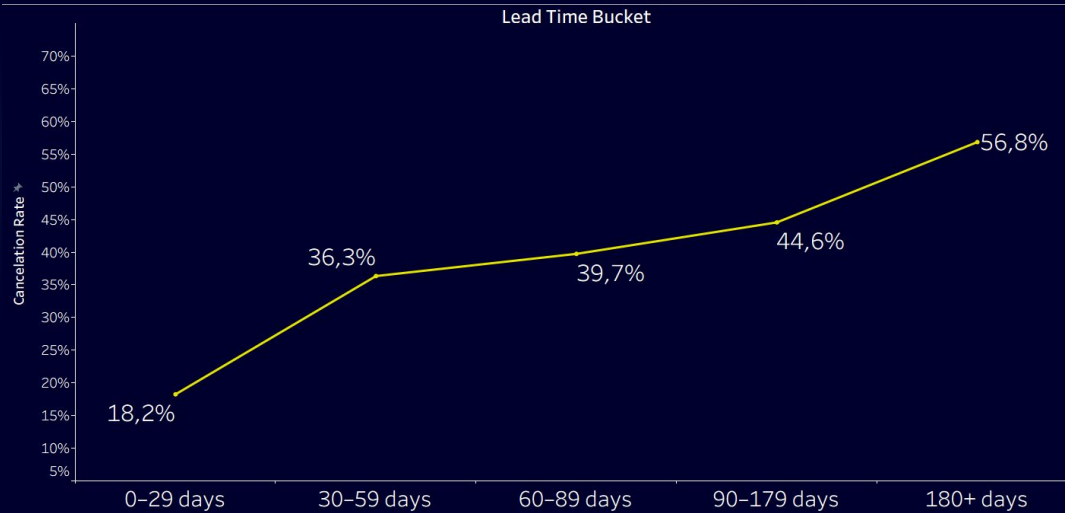
TA/TO accounts for ~94% of non-refundable bookings, while cancellation rates exceed 94% across all distribution channels.

Non-Refundable Bookings Have 100% Cancellation Among Contract and Transient Customers

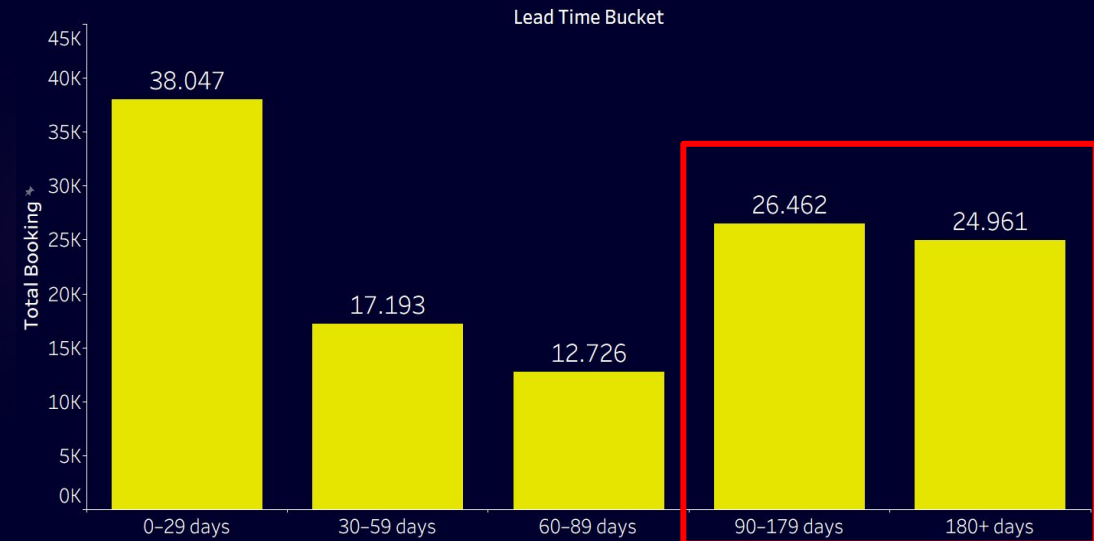
Hotel	Depo..	Customer Type	Cancellation Rate	Total Booking	Total Cancelled
City	Non Refund	Contract	100,0%	540	540
		Transient	100,0%	11.290	11.290
		Transient-Party	97,7%	1.038	1.014
Resort	Non Refund	Contract	100,0%	4	4
		Transient	100,0%	1.619	1.619
		Transient-Party	28,1%	96	27

- Contract and Transient bookings: **100% cancellation** for non-refundable bookings across both hotels
- Transient-Party: cancellation **differs sharply – 97.7% City vs. 28.1% Resort.**

Longer Lead Times Are Associated With Higher Cancellation Risk



Cancellation rates increase consistently with lead time, from **18.2%** for short-term bookings to **56.8%** for bookings made 180+ days in advance



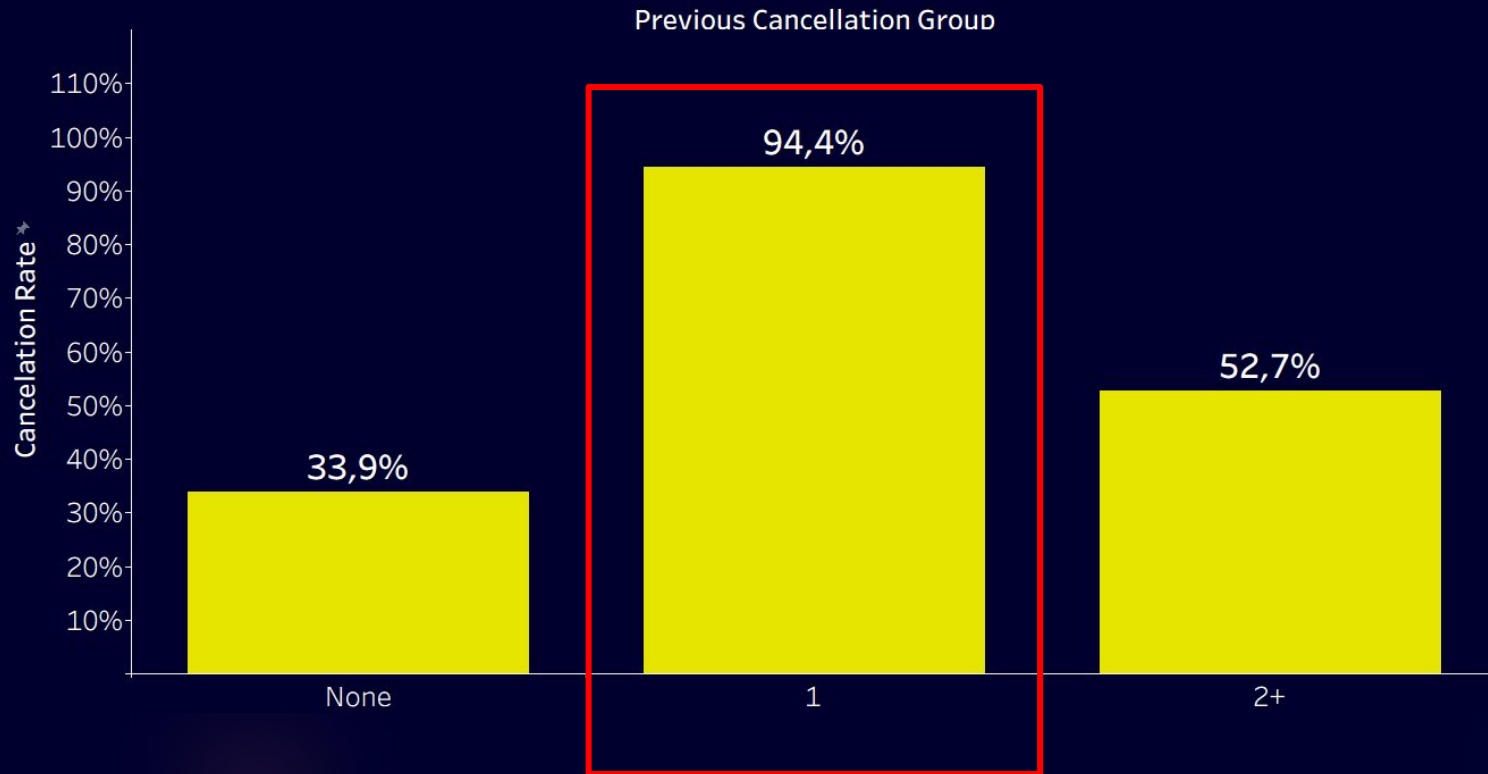
Short-lead bookings have the highest volume, but **51.4K bookings** are made **90+ days** in advance, creating a **substantial pool of long-lead reservations**.

Long Lead-Time TA/TO Bookings Show Higher Cancellation Risk

Lead Time	Distribution Channel			
	Corporate	Direct	GDS	TA/TO
0–29 days	13,9% 4.639 Bookings 645 Cancelled	11,6% 8.152 Bookings 949 Cancelled	15,4% 143 Bookings 22 Cancelled	21,2% 25.109 Bookings 5.322 Cancelled
30–59 days	43,8% 737 Bookings 323 Cancelled	22,4% 1.738 Bookings 389 Cancelled	21,1% 38 Bookings 8 Cancelled	37,7% 14.680 Bookings 5.528 Cancelled
60–89 days	41,7% 338 Bookings 141 Cancelled	21,3% 1.075 Bookings 229 Cancelled	42,9% 7 Bookings 3 Cancelled	41,4% 11.306 Bookings 4.684 Cancelled
90–179 days	50,5% 372 Bookings 188 Cancelled	27,0% 2.349 Bookings 635 Cancelled	75,0% 4 Bookings 3 Cancelled	46,2% 23.736 Bookings 10.963 Cancelled
180+ days	29,9% 591 Bookings 177 Cancelled	26,7% 1.330 Bookings 355 Cancelled	100,0% 1 Bookings 1 Cancelled	59,3% 23.039 Bookings 13.655 Cancelled

TA/TO bookings with a 180+ day lead time have a 59.3% cancellation rate, compared with 21.2% for bookings made within 0–29 days. With 23,039 bookings in the 180+ day segment, **this represents 13,655 cancellations**

Previous Cancellation History Signals Strong Future Cancellation Risk



Insight:

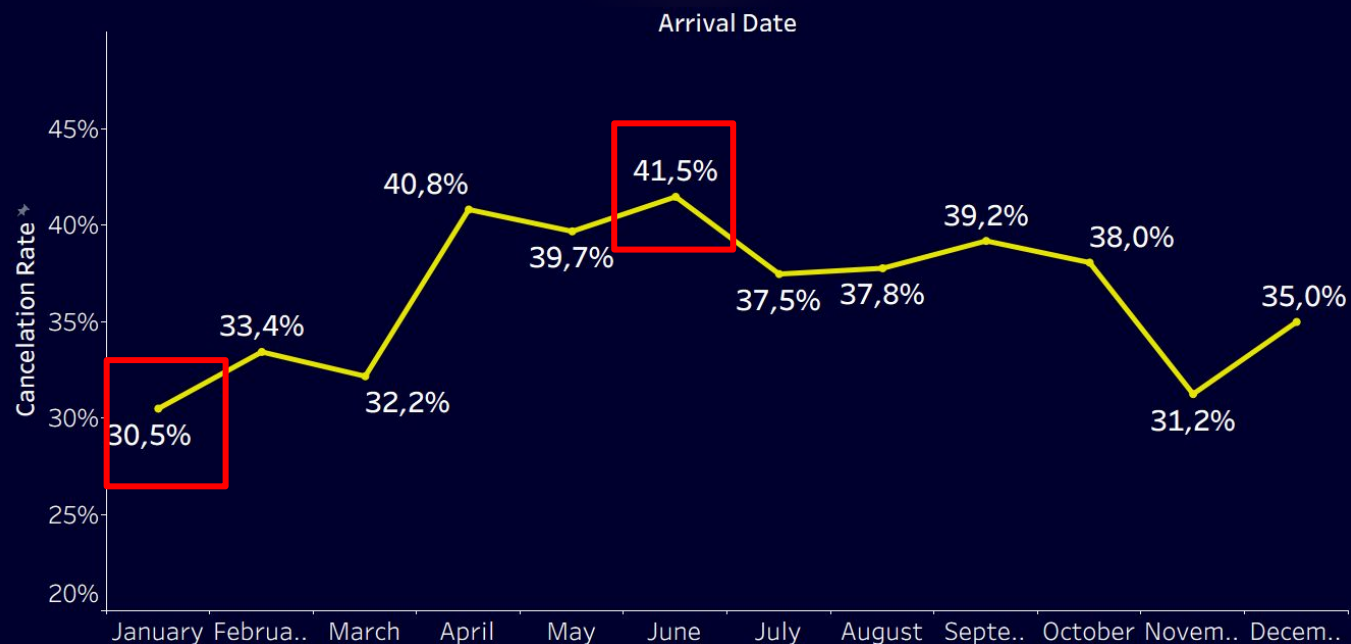
Previous cancellation history is strongly associated with cancellation risk. Customers with exactly **one previous cancellation** have a **94.4%** cancellation rate, compared with 33.9% for customers with no previous cancellations.

Cancellation Rate by Previous Cancellation History

Previous Cancellation History	Customer Type			
	Contract	Group	Transient	Transient-Party
None	10,9% 3.158 Bookings 344 Cancelled	7,9% 558 Bookings 44 Cancelled	38,9% 86.021 Bookings 33.429 Cancelled	19,3% 23.168 Bookings 4.465 Cancelled
2+	100,0% 1 Bookings 1 Cancelled	0,0% 2 Bookings 0 Cancelled	46,7% 351 Bookings 164 Cancelled	79,7% 79 Bookings 63 Cancelled
1	100,0% 917 Bookings 917 Cancelled	88,2% 17 Bookings 15 Cancelled	90,1% 3.241 Bookings 2.921 Cancelled	99,2% 1.876 Bookings 1.861 Cancelled

Bookings with one previous cancellation show cancellation rates above 88% across all customer types, compared with substantially lower rates among customers with no previous cancellations.

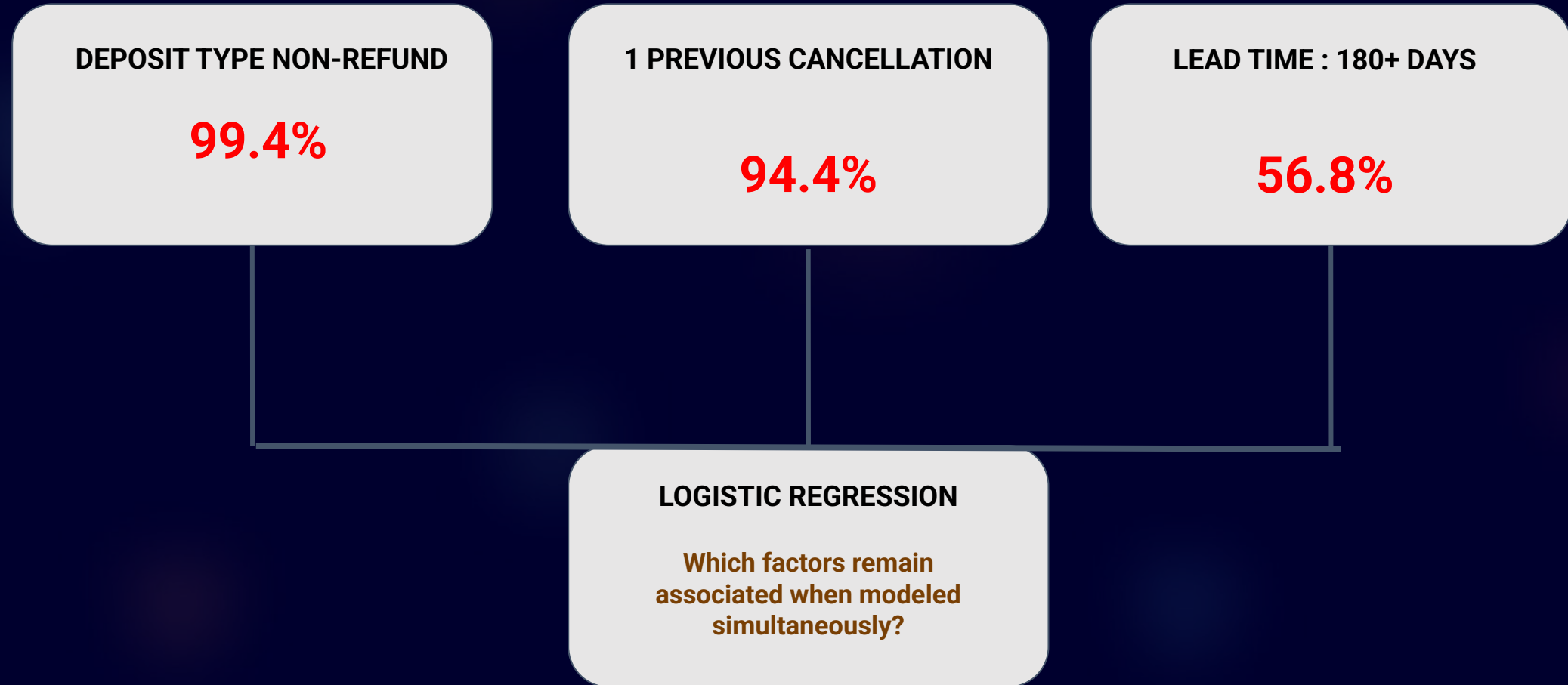
Cancellation rate based on the scheduled arrival month



Arrival month
↓
When guests are expected to stay
↓
Seasonal demand / occupancy
pattern

Cancellation is highest **June (41.5%)** and lowest in **January (30.5%)**, but the ~11 pp range suggests **seasonality is a supporting factor rather than a primary driver.**

EDA Findings to Logistic Regression



Top 4 Factors Associated with Cancellation — Logistic Regression

Top 4 Output Based on Logistic Regression

Features	coefficient	odds_ratio
deposit_type_Non Refund	4.946015	140.613525
previous_cancellations	2.290354	9.878434
customer_type_Transient	0.756443	2.130684
lead_time	0.462763	1.588456

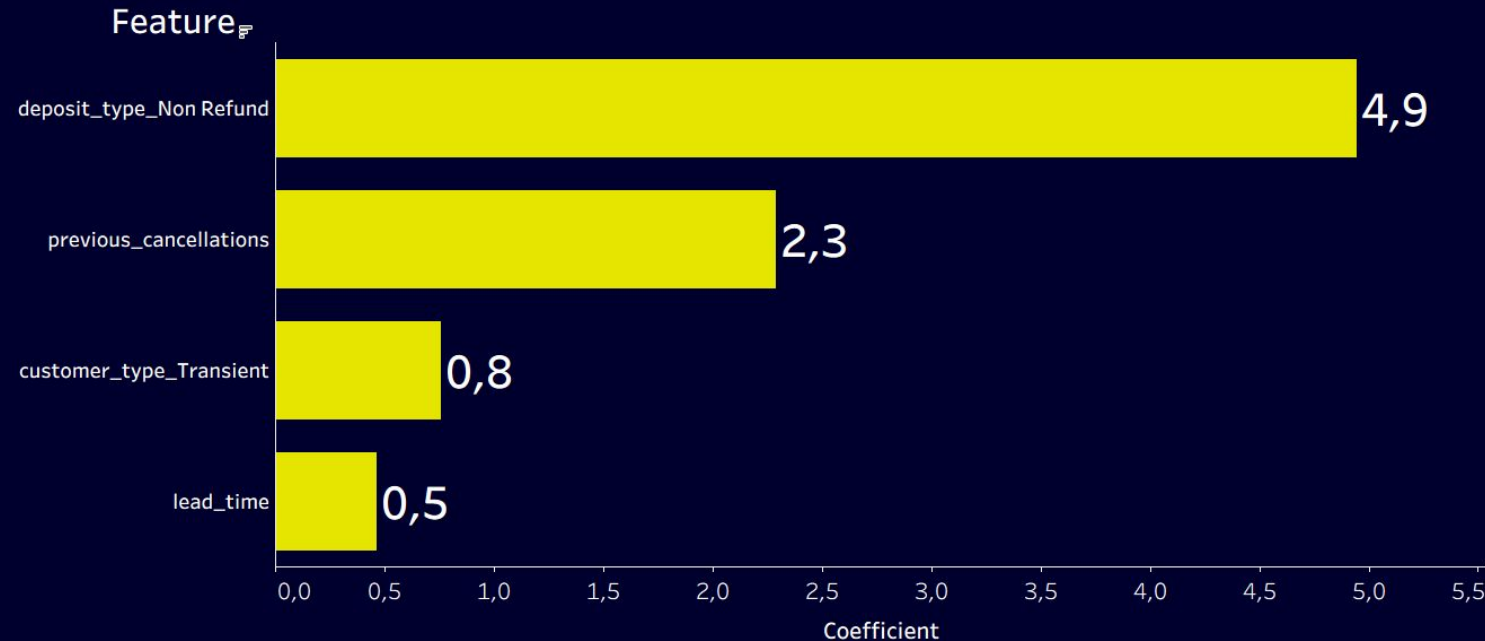
Model Evaluation — Test Set

Accuracy	Precision	Recall	F1	ROC-AUC
79.9%	86.0%	55.0%	67.0%	84.1%

Odds Ratio compares the odds of cancellation between categories or for a one-unit increase in a predictor. $OR > 1$ indicates higher cancellation odds.

Model performance: 84.1% ROC-AUC indicates good ability to distinguish cancelled vs. non-cancelled bookings, while 55.0% recall means the model misses some actual cancellations.

Logistic Regression Identifies Key Cancellation Risk Factors



Deposit type and previous cancellation history are the strongest factors associated with booking cancellation.

+4.9
Non-Refund Deposit
Strongest positive association

+2.3
Previous Cancellations
Higher cancellation tendency among customers with prior cancellations

+0.8
Transient's Customer
Transient's customer is associated with higher cancellation odds

+0.50
Lead Time
Longer lead time is associated with higher cancellation odds

Recommendation and Insight

1 90-Day Lead-Time Monitoring Rule

Flag bookings made 90+ days before arrival for additional confirmation monitoring

Business Impact: Improve demand forecasting and reduce occupancy uncertainty.

2 Review TA/T0 Partner Booking & Confirmation Processes

Monitor partner performance monthly and Escalate partners whose cancellation performance deteriorates

Business Impact: Reduce cancellations and improve occupancy predictability.

3 Build a Cancellation-Risk Flag System

Use the logistic regression model to classify bookings by cancellation risk and prioritize high-risk bookings for monitoring.

Business Impact: Prioritize resources and improve occupancy forecasting.

Dashboard Overview

Hotel Booking Cancellation Performance

Deposit Type

[All]

Hotel

[All]

Distribution Channel

[All]

▼

Arrival Date

[All]

0

< >

Total Booking

119,389

Cancelled

44,224

Avg Lead Time

104D

Cancellation Rate

37%

Cancellation by Arrival Date



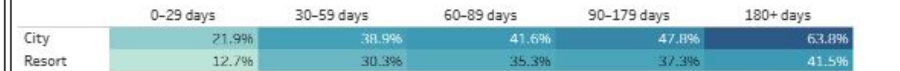
Cancellation rate by lead time



Cancellation Rate by hotel



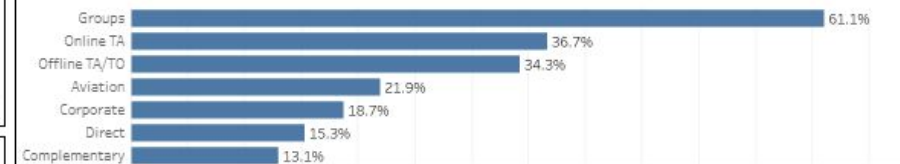
Cancellation Rate by Hotel & Lead Time



Cancellation Rate by Customer Type



Cancellation Rate by Market Segment



Cancellation Rate by Deposit Type



APPENDIX

Data Validation & Problem Framing

Dataset Overview

The dataset provides data at the granularity of individual bookings/reservations. Each row contains comprehensive information about a reservation, including customer details, booking characteristics, stay specifics, pricing, and the final reservation status. The dataset comprises 119,390 bookings across 32 columns. Key data points include booking status, arrival dates, customer characteristics, stay duration, room information, booking channels, cancellation history, and Average Daily Rate (ADR).

KPI Feasibility

- Cancellation rate

(cancelled booking ÷ total booking x 100)

- Total Bookings

- Average Lead Time

(Total Lead Time ÷ Number of records)

Feasibility Awareness

The current dataset has limitations that impact certain business analyses. It does not provide actual revenue figures, specific reasons for cancellations, detailed customer demographics, or explicit hotel names. These limitations prevent answering critical questions such as:

- Why did customers cancel their bookings?
- Which specific hotel location performs best?

Initial Business question:

- What factors are associated with hotel cancellations?
- Which booking channels generate the most bookings?

Data Cleaning

RAW DATA

Fixing Data Types

Imputing Missing
Value

Fixing
Inconsistency/typo

Feature Creation

Data Type Correction : 7 variables addressed

Date → datetime

Agent / Company → string

Binary variables → boolean

Children → integer

Missing Values: 129,425 missing values identified across 4 columns

country → "Unknown"

agent → retained as missing

company → removed

children → 0

Data Consistency: Country code standardized

CN → CHN

Feature Engineering: Created arrival_date

Year + Month + Day → YYYY-MM-DD

Reasoning for Imputation

1. country — Fill with "Unknown"
country is categorical and the actual country cannot be reliably inferred, "Unknown" preserves the records without introducing a false country.
2. agent — Keep as NaN
cannot determine whether the agent was genuinely absent or simply not recorded, so retaining NaN preserves that uncertainty.
3. children — Fill with 0
children represents the number of children included in the booking, 0 is a reasonable assumption when no children are recorded.
4. company — Drop
112,593 records (~94.3%) are missing. Keeping it could introduce unnecessary noise and complexity.

Reasoning for not deleting duplication:

31,994 exact duplicate rows were identified, but were retained because the dataset does not contain a unique booking identifier.

Outliers Assessment and Logical anomalies

lead time

outliers:3005

Lower bound: -195.0

Upper bound: 373.0

adr

Lower fence: -15.774999999999991

Upper fence: 211.065

Outliers: 3793

previous_cancellations

Lower fence: 0.0

Upper fence: 0.0

Outliers: 6484

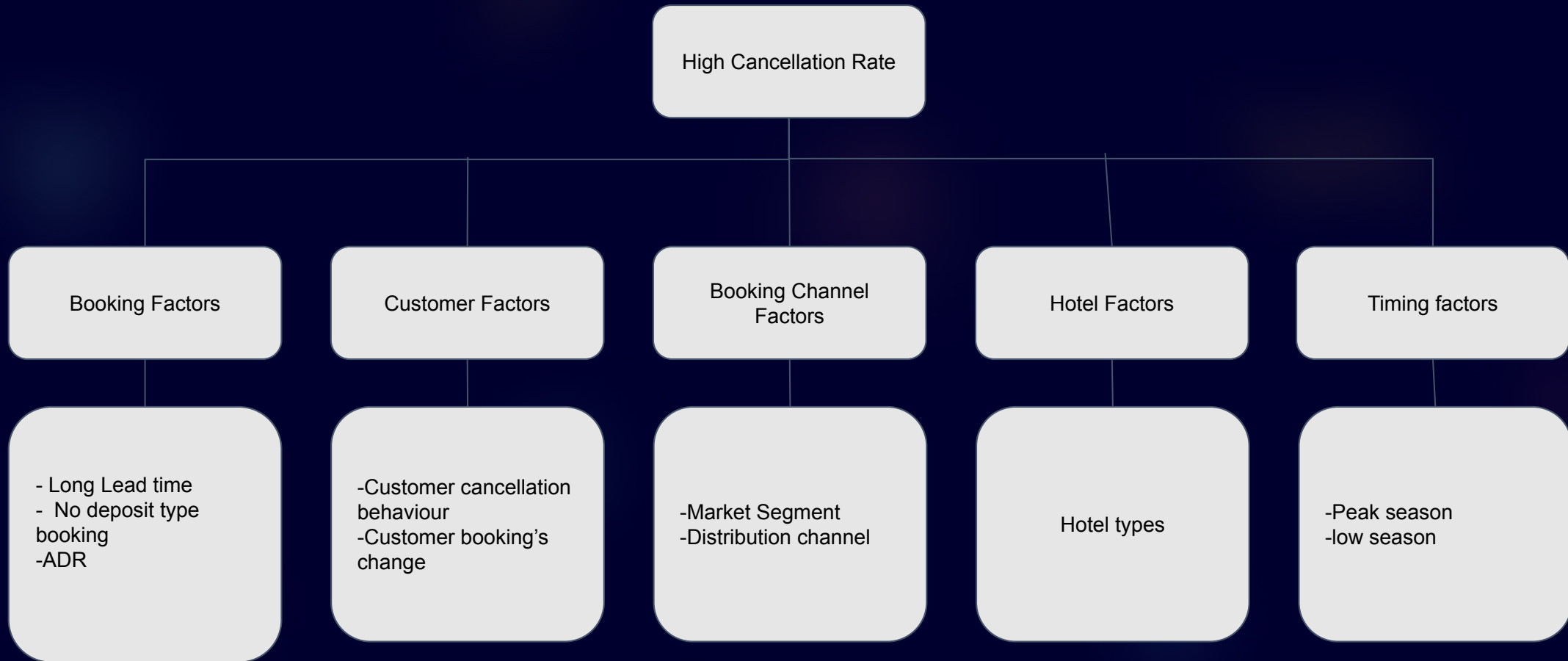
Logical anomalies found

Zero-guest records: 180 records had no adults, children, or babies recorded. No clear data-quality error could be established, so these records were retained to avoid removing potentially valid booking records.

Negative ADR: 1 record had an impossible ADR of -6.38. The record was removed as a confirmed data-quality error, with negligible impact on the analysis.

*IQR outliers were retained because an outlier does not necessarily indicate a data error. Extreme values may represent legitimate booking behavior and could contain important information about cancellation patterns

Root Cause Analysis



Key findings

Finding	Business Implication
Longer lead time → higher cancellation	180+ day bookings reach 56.8% cancellation , increasing demand uncertainty
Cancellation varies by segment	Groups reach 61.1% , showing the need for segment-level risk monitoring
City Hotel has higher cancellation	41.7% vs. 27.8% at Resort Hotel, creating greater forecasting uncertainty
Non-Refund shows an extreme pattern	99.4% cancellation warrants further investigation of booking policy and data

EDA Summary – Factors Associated With Cancellation

Strongest Factors

- **Deposit Type:** Non-refundable bookings show an exceptionally high **99.4% cancellation rate**.
- **Segment Composition:** Non-refundable bookings are heavily concentrated in **Groups and Offline TA/TO**, which also show exceptionally high cancellation rates, contributing substantially to the overall Non Refund cancellation rate.
- **Lead Time:** Cancellation risk rises sharply with longer lead times, reaching **~70% for bookings made 365+ days in advance**.
- **Previous Cancellations:** Customers with previous cancellations have a **94.4% cancellation rate**, indicating strong customer-level cancellation behavior.

Key Segment & Channel Factors

- **Market Segment:** Groups have the highest cancellation rate at **61.1%**.
- **Distribution Channel:** TA/TO has the highest cancellation rate among major channels at **41.0%** and contributes the largest cancellation volume.
- **Hotel Type:** City Hotel has a higher cancellation rate than Resort Hotel (**41.7% vs 27.8%**).

Supporting Factor

- **Seasonality:** Cancellation rates vary from **~30.5% in January to ~41.5% in June**, indicating a moderate seasonal effect.

Hypotheses Not Supported

- **ADR:** No meaningful increase in cancellation rate as ADR increases.
- **Booking Changes:** More booking changes do not correspond to higher cancellation.

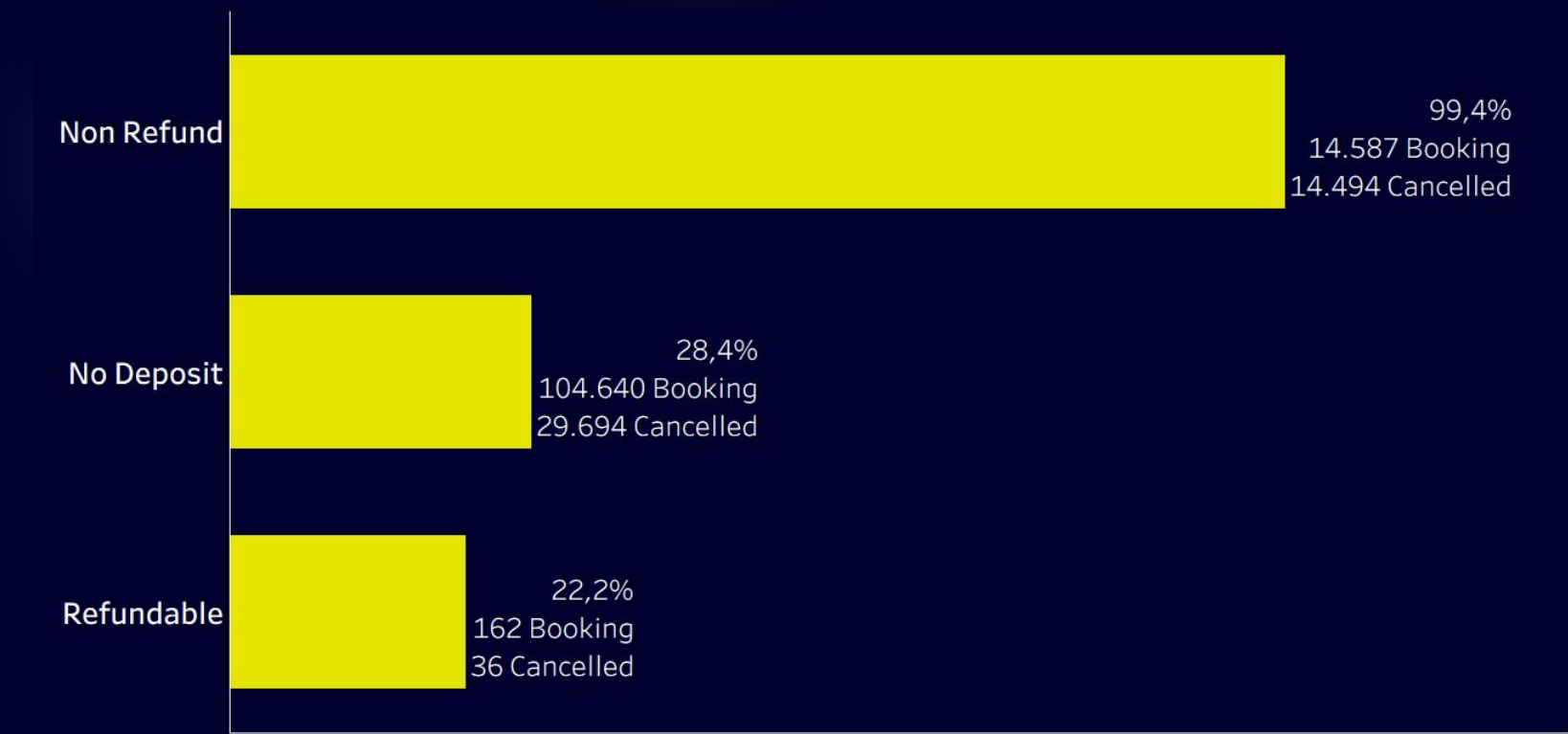
Key Insight :

Cancellation is most strongly associated with booking conditions and customer history—particularly deposit type, lead time, and previous cancellations—while booking channel and customer segment help identify where cancellation risk is concentrated.

Stakeholder Identification(DARCI)

Decision maker	General Manager
Accountable	Revenue Manager
Responsible	Data Analyst Team
Consulted	Front office Manager, Operation Manager
Informed	Operation Team

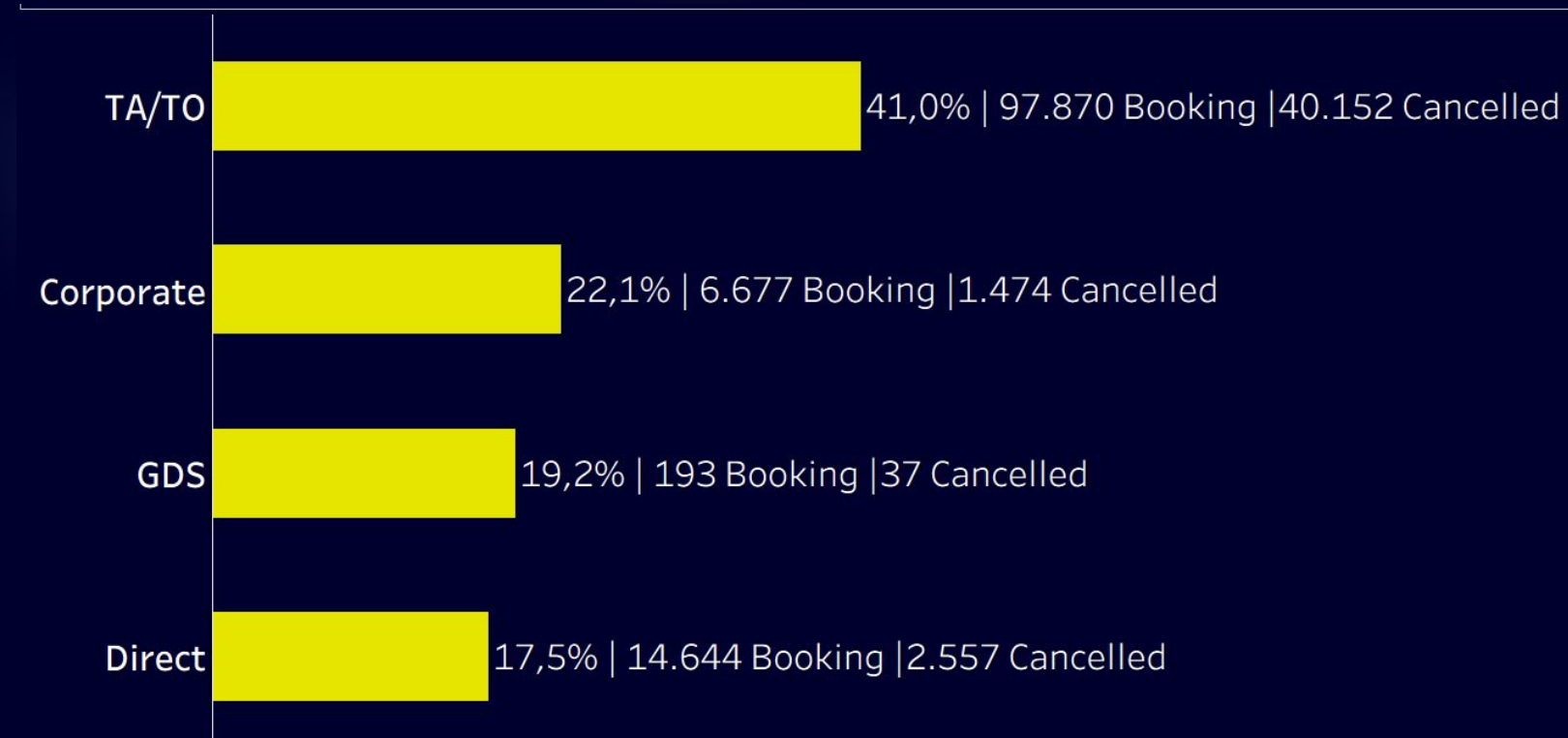
Cancellation Rate by Deposit Type



 **Insight:**

Non Refund bookings have a **99.4% cancellation rate** — nearly all bookings in this category are canceled.

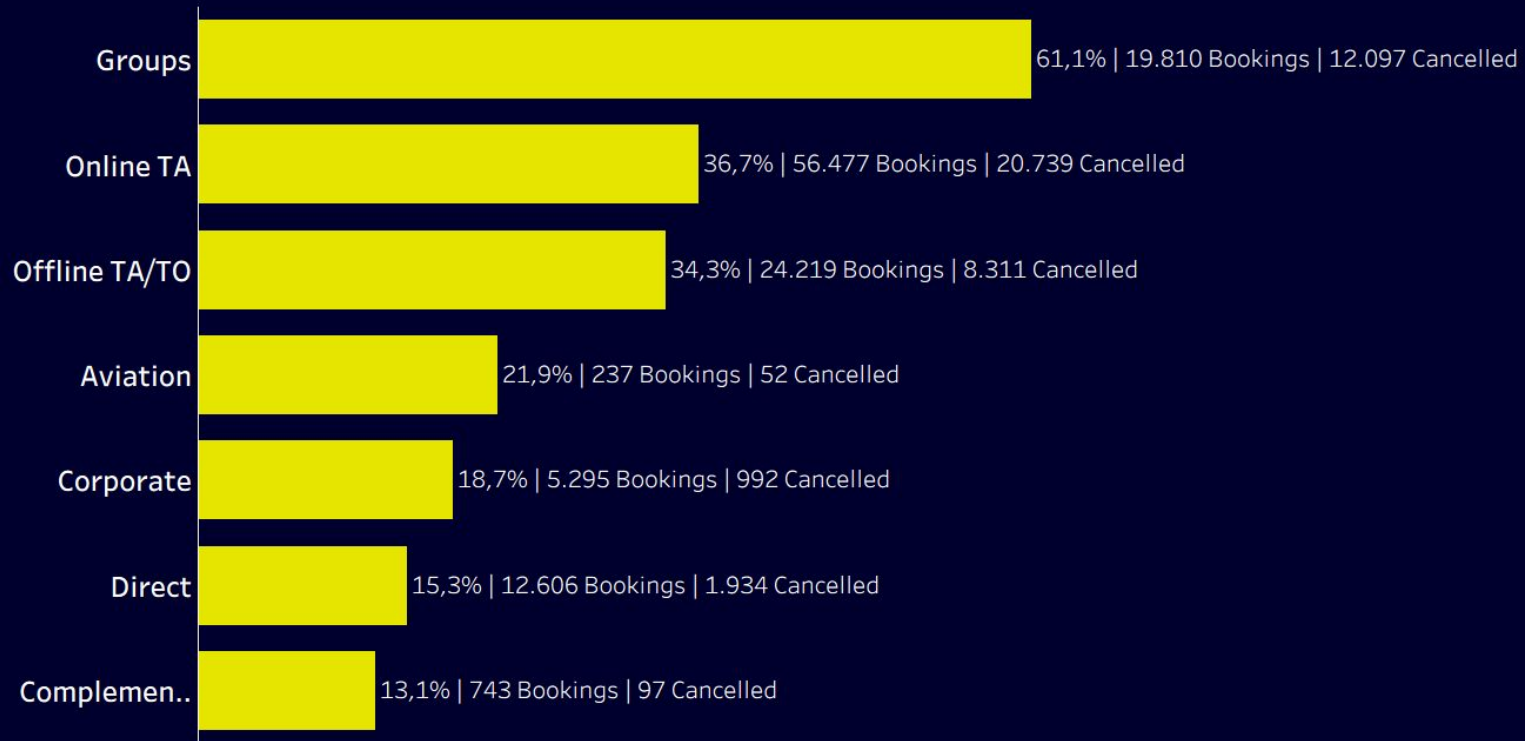
Cancellation Rate by Distribution Channel



Insight:

TA/TO bookings show the highest cancellation rate among major distribution channels (**41.0%**) and account for **40,152** canceled records

Cancellation Rate by Market Segment

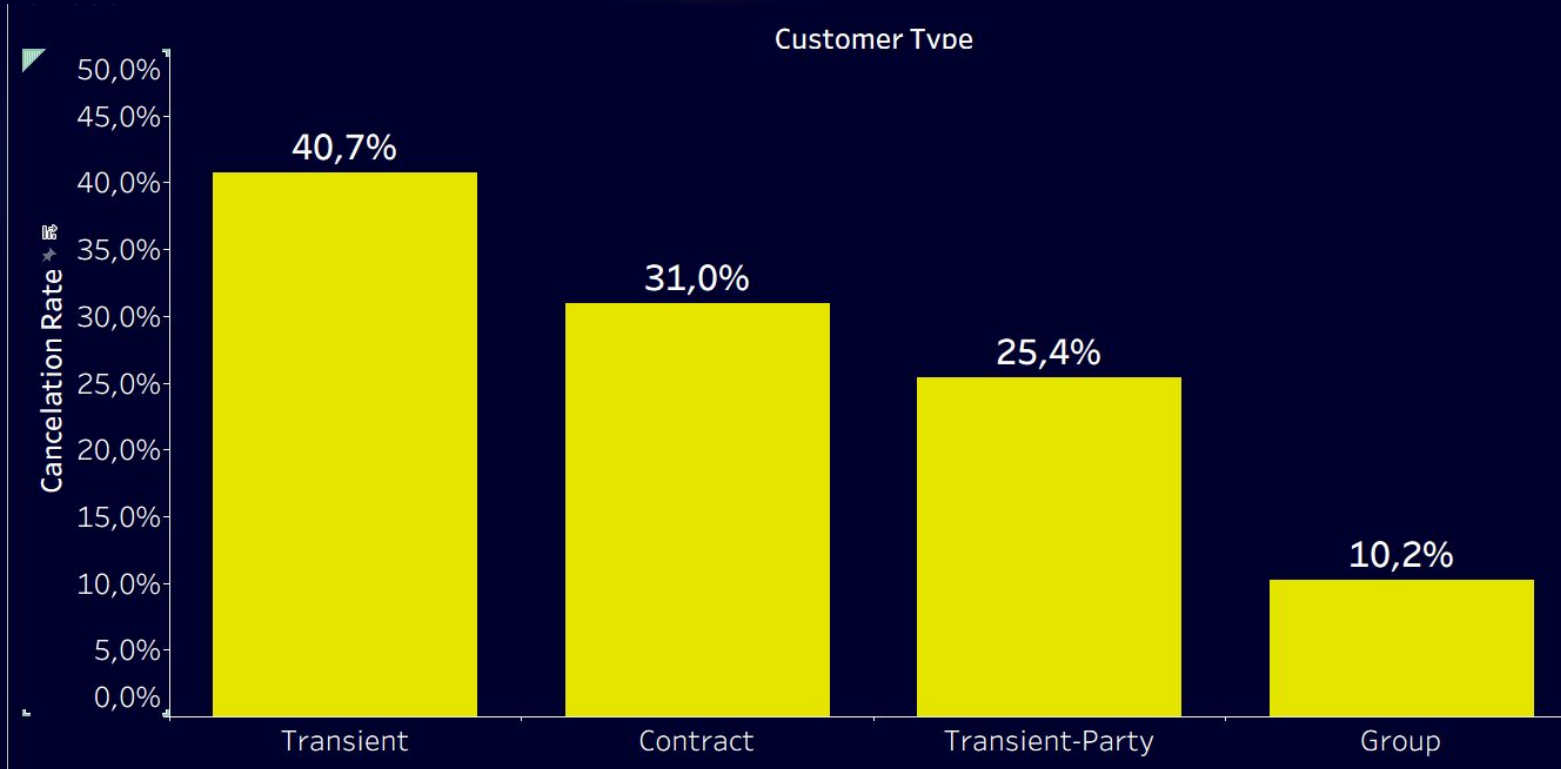


Insight:

Groups have the highest cancellation rate (**61.1%**),

Online TA contributes the most canceled records (**20,739**) due to its high booking volume.

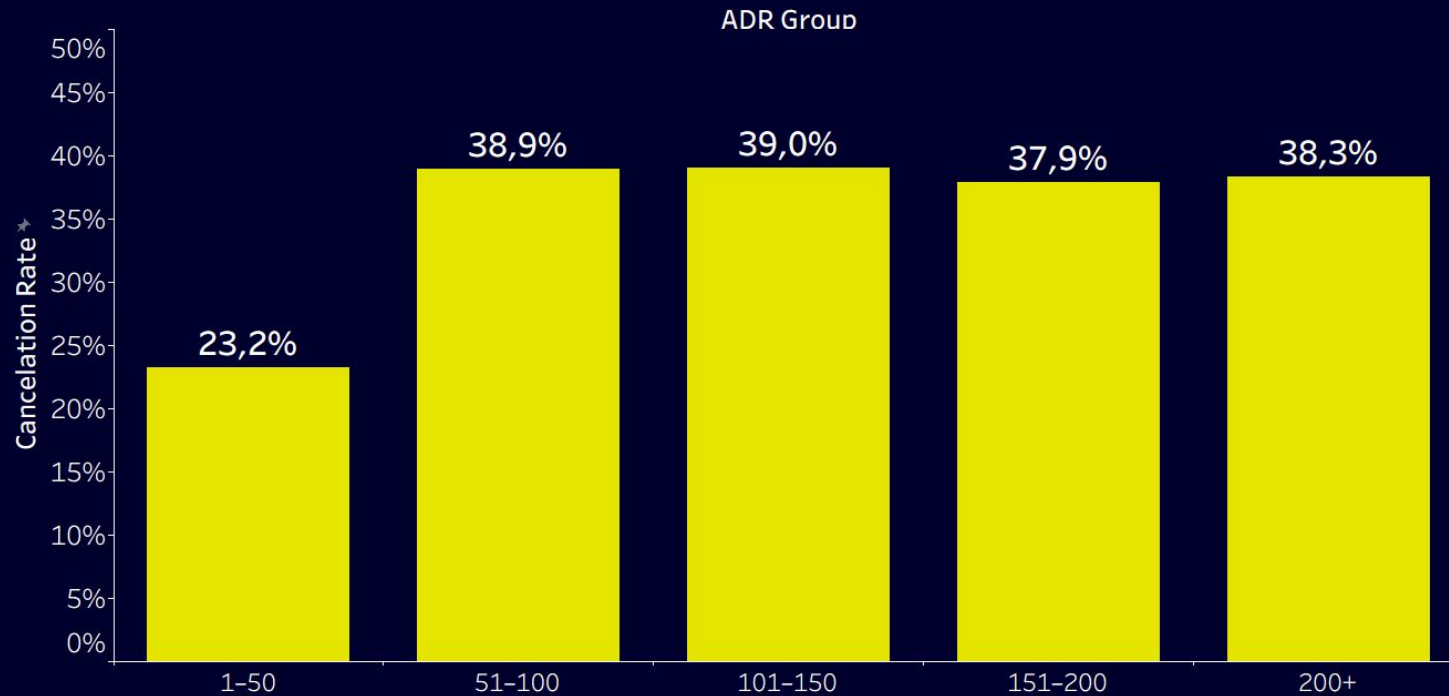
Cancellation Rate by Customer Type



Insight:

Transient bookings have the highest cancellation rate at 40.7%, followed by Contract (31.0%) and Transient-Party (25.4%), while Group bookings have the lowest rate at 10.2%.

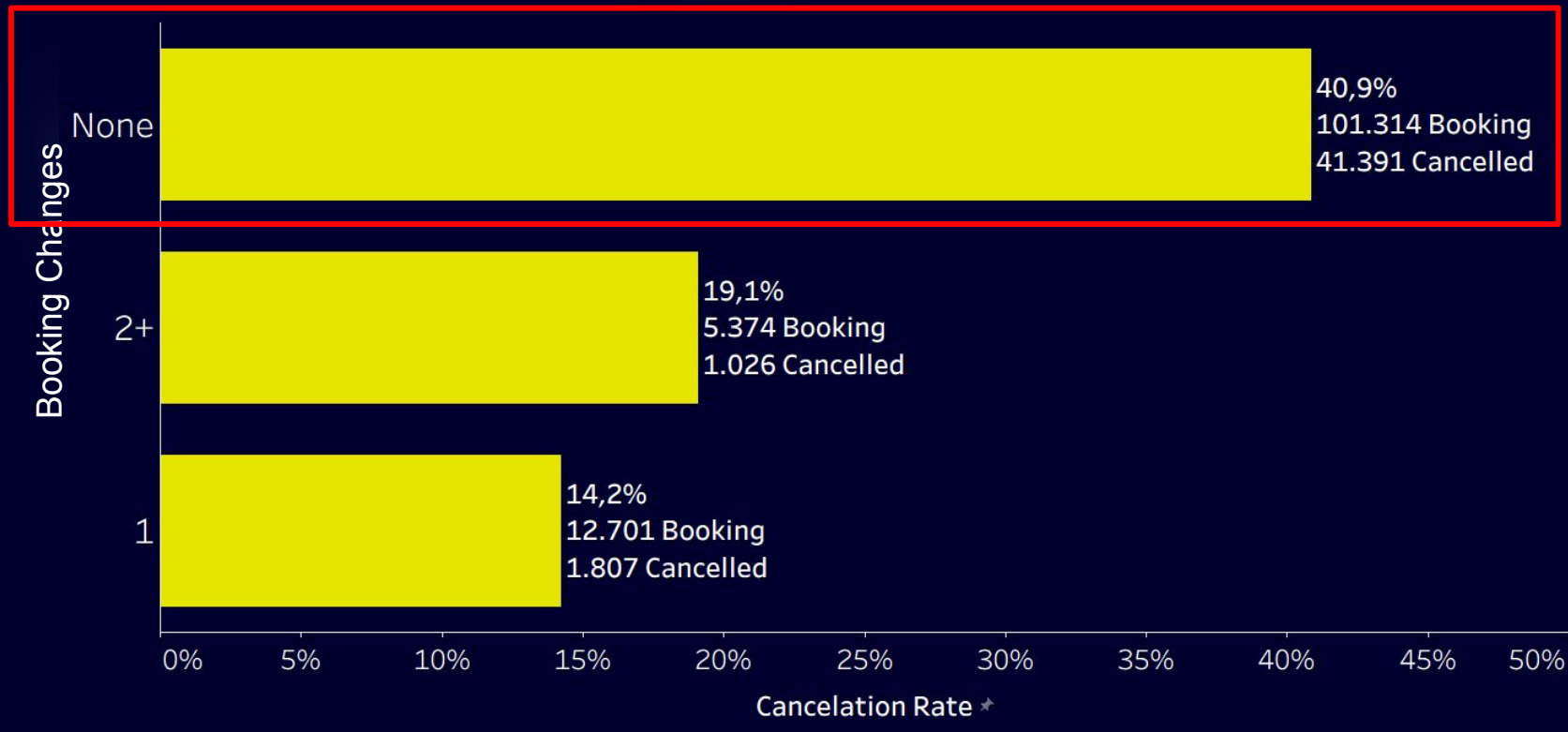
Cancellation Rate by ADR



Insight:

ADR above 50 shows little variation in cancellation rates (38.0–38.9%), suggesting that higher ADR is not a major driver of cancellation.

Cancellation Rate by How many times Booking Changes



Insight:

Bookings with no changes have a 40.9% cancellation rate, compared with 14.2% for bookings with one change, indicating that booking changes are not positively associated with cancellation

Logistic Regression Workflow

Logistic Regression Variable

Dependent Variable:
is_cancelled

Independent Variable

Booking characteristics: Lead Time, Booking Changes, Deposit Type

Customer characteristics: Customer Type, Repeat Guest, Previous Cancellation History

Stay & pricing: ADR, Total Nights, Total Guests

Booking channel: Hotel, Meal, Distribution Channel, Room Type

Other: Waiting List, Parking Spaces, Special Requests

Precision, Recall and F1-Score are reported for the Cancelled class.

Logistic Regression Workflow

1. Define Target & Predictors

Set `is_canceled` as the target and use booking, customer, stay, pricing, and channel characteristics as predictors.

2. Select & Clean Variables

Remove irrelevant/redundant variables, address multicollinearity using VIF, and remove logically invalid records.

3. Split the Data

Split the dataset into **80% training and 20% testing**, using stratification to preserve the cancellation ratio.

4. Encode Categorical Variables

Apply **One-Hot Encoding** with the first category dropped as the reference category.

5. Scale Numerical Variables

Apply **StandardScaler** to numerical predictors so variables are on comparable scales.

6. Train Logistic Regression Model

Fit the logistic regression model using the processed training data and estimate each predictor's coefficient and **odds ratio**.

7. Predict & Evaluate

Generate cancellation predictions and probabilities, then evaluate the model using **Accuracy, Precision, Recall, F1-score, and ROC-AUC**.

Goal: Identify which booking characteristics are most strongly associated with cancellation and estimate cancellation likelihood.